





## Qualification Pack

### Contents

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| LFS/Q0314: Chemical Laboratory Technology .....                                | 3  |
| <i>Brief Job Description</i> .....                                             | 3  |
| Applicable National Occupational Standards (NOS) .....                         | 3  |
| <i>Compulsory NOS</i> .....                                                    | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                | 3  |
| LFS/N0354: Work organization and management .....                              | 5  |
| LFS/N0355: Communication and interpersonal skills .....                        | 10 |
| LFS/N0356: Techniques, procedures, and methods .....                           | 15 |
| LFS/N0357: Data processing and record keeping .....                            | 20 |
| LFS/N0358: Analysis, interpretation, and evaluation .....                      | 24 |
| LFS/N0359: Problem solving through the application of scientific methods ..... | 28 |
| LFS/N0360: Trends in applied Chemistry .....                                   | 32 |
| Assessment Guidelines and Weightage .....                                      | 35 |
| <i>Assessment Guidelines</i> .....                                             | 35 |
| <i>Assessment Weightage</i> .....                                              | 36 |
| Acronyms .....                                                                 | 38 |
| Glossary .....                                                                 | 39 |



## Qualification Pack

### LFS/Q0314: Chemical Laboratory Technology

#### Brief Job Description

The work of laboratory chemical analyst is the basis of product quality in many industries. Chemical analysis is necessary to control the compliance of the properties of raw materials, intermediate stages of the technological process, and finished products with current standards. The laboratory chemical analyst performs laboratory analyses, tests, measurements aimed at determining The qualitative chemical composition of a substance The quantitative ratio of chemical elements and compounds within them Processing of the obtained data, Reporting results of analyses and Other types of laboratory work according with the requirements of standards and specifications. The scope and complexity of the tests performed depend on the type of tasks set by the employer

#### Personal Attributes

Laboratory chemical analysts should be ready to determine the optimal tools and methods for the analysis of various natural and synthetic materials, to perform qualitative and quantitative tests using modern chemical, bioanalytical and physicochemical analytical methods. They should be able to act logically and systematically, complying with sanitary and hygienic requirements and occupational safety and health standards.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [LFS/N0354: Work organization and management](#)
2. [LFS/N0355: Communication and interpersonal skills](#)
3. [LFS/N0356: Techniques, procedures, and methods](#)
4. [LFS/N0357: Data processing and record keeping](#)
5. [LFS/N0358: Analysis, interpretation, and evaluation](#)
6. [LFS/N0359: Problem solving through the application of scientific methods](#)
7. [LFS/N0360: Trends in applied Chemistry](#)

#### Qualification Pack (QP) Parameters

|        |               |
|--------|---------------|
| Sector | Life Sciences |
|--------|---------------|



## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sub-Sector</b>                                         | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology                                                                                                                                                                                                                                                                                   |
| <b>Occupation</b>                                         | Quality                                                                                                                                                                                                                                                                                                                                                   |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                                                                                     |
| <b>NSQF Level</b>                                         | 4.5                                                                                                                                                                                                                                                                                                                                                       |
| <b>Credits</b>                                            | 17                                                                                                                                                                                                                                                                                                                                                        |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/2131.1300                                                                                                                                                                                                                                                                                                                                        |
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>Pursuing 1st year of UG (B. Pharma. /UG with Chemistry Subject/ B.Tech. in relevant field))<br/>OR<br/>Pursuing 2nd year of 2-year diploma after 12th (in relevant field)<br/>OR<br/>Pursuing 3rd year of 3-year diploma after 10th (in relevant field)<br/>OR<br/>12th grade Pass (with Science Subject) with 1 Year of experience relevant field</p> |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                           |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                        |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                  |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                        |
| <b>Next Review Date</b>                                   | 08/02/2026                                                                                                                                                                                                                                                                                                                                                |
| <b>NSQC Approval Date</b>                                 | 08/02/2024                                                                                                                                                                                                                                                                                                                                                |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                       |
| <b>Reference code on NQR</b>                              | QG-4.5-LS-01815-2024-V1-LSSSC                                                                                                                                                                                                                                                                                                                             |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                                                                         |



## Qualification Pack

### LFS/N0354: Work organization and management

#### Description

This Nos is about Work organization and management

#### Scope

The scope covers the following :

- this scope of this NOS is Work organization and management

#### Elements and Performance Criteria

##### *work organization and management*

To be competent, the user/individual on the job must be able to:

- PC1.** Maintain personal health and safety at all times including through personal protective clothing and equipment
- PC2.** Perform work taking into account relevant regulations, norms, quality, safety and environmental standards
- PC3.**
  - Implement safety data sheets and the measures and procedures derived from them for
  - • Handling, maintaining, and repairing laboratory devices, apparatus, and equipment
  - • Handling, maintaining, and disposing of/recycling chemicals used in laboratories
- PC4.** Follow risk management systems, proactively
- PC5.** Maintain good housekeeping
- PC6.** Order and inventory materials to maintain supplies, within budget and budgetary procedures
- PC7.** Ensure electronic equipment is fit for purpose
- PC8.** Inspect structures and materials for their condition and usability
- PC9.** Work independently, taking responsibility for initiating and completing tasks within the parameters of the work role
- PC10.** Estimate the requirements of a piece of work in terms of time, costs, resources, and materials needed for completion.
- PC11.** Develop specific goals and plans to prioritize, organize, and accomplish work, relative to set objectives and targets
- PC12.** Investigate alternative means of resolving delays
- PC13.** Adjust activities as required, keeping relevant others informed

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The internal and external regulatory environment for the sector
- KU2.** The internal business environment including one's personal role, ethical practice, and codes of conduct
- KU3.** Health and safety legislation, regulations, and best practice



## Qualification Pack

- KU4.** Scientific principles for laboratory-based activities
- KU5.** Principles for work planning, scheduling, organization, and completion
- KU6.** The theoretical basis for applied chemistry including how to apply physical, organic, and inorganic chemistry to laboratory work
- KU7.** Principles and methods for the safe disposal or recycling of chemicals and chemically related substances

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of EHS protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ EHS Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>work organization and management</i>                                                                                                                                                                                                                                                                                                   | <b>30</b>    | <b>70</b>       | -             | -          |
| <b>PC1.</b> Maintain personal health and safety at all times including through personal protective clothing and equipment                                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC2.</b> Perform work taking into account relevant regulations, norms, quality, safety and environmental standards                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> <ul style="list-style-type: none"> <li>Implement safety data sheets and the measures and procedures derived from them for</li> <li>• Handling, maintaining, and repairing laboratory devices, apparatus, and equipment</li> <li>• Handling, maintaining, and disposing of/recycling chemicals used in laboratories</li> </ul> | -            | -               | -             | -          |
| <b>PC4.</b> Follow risk management systems, proactively                                                                                                                                                                                                                                                                                   | -            | -               | -             | -          |
| <b>PC5.</b> Maintain good housekeeping                                                                                                                                                                                                                                                                                                    | -            | -               | -             | -          |
| <b>PC6.</b> Order and inventory materials to maintain supplies, within budget and budgetary procedures                                                                                                                                                                                                                                    | -            | -               | -             | -          |
| <b>PC7.</b> Ensure electronic equipment is fit for purpose                                                                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC8.</b> Inspect structures and materials for their condition and usability                                                                                                                                                                                                                                                            | -            | -               | -             | -          |
| <b>PC9.</b> Work independently, taking responsibility for initiating and completing tasks within the parameters of the work role                                                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC10.</b> Estimate the requirements of a piece of work in terms of time, costs, resources, and materials needed for completion.                                                                                                                                                                                                        | -            | -               | -             | -          |
| <b>PC11.</b> Develop specific goals and plans to prioritize, organize, and accomplish work, relative to set objectives and targets                                                                                                                                                                                                        | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> Investigate alternative means of resolving delays               | -            | -               | -             | -          |
| <b>PC13.</b> Adjust activities as required, keeping relevant others informed | -            | -               | -             | -          |
| <b>NOS Total</b>                                                             | <b>30</b>    | <b>70</b>       | -             | -          |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0354                                                               |
| <b>NOS Name</b>            | Work organization and management                                        |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 1.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |



## Qualification Pack

### LFS/N0355: Communication and interpersonal skills

#### Description

This NOS is about Communication and interpersonal skills

#### Scope

The scope covers the following :

- The scope of this NOS is Communication and interpersonal skills

#### Elements and Performance Criteria

##### *Communication and interpersonal skills*

To be competent, the user/individual on the job must be able to:

- PC1.** Establish and maintain interpersonal relations
- PC2.** Work and interact with others including within teams
- PC3.** Provide technical support to chemists or other specialists
- PC4.** Communicate using a full range of techniques for speaking, writing, body language, and active listening, for formal and informal purposes
- PC5.** Use specialist terminology including where these are in another language
- PC6.** Obtain information from all relevant sources, citing sources as required
- PC7.**
  - Read and apply the content of technical documents relating to
    - ◦ Analyses
    - ◦ Formulations
    - ◦ Procedural instructions
    - ◦ Specifications
    - ◦ Diagrams
- PC8.** Listen actively, asking questions appropriately for full understanding
- PC9.** Use laboratory information and laboratory management systems, both digital and paper based
- PC10.** Order information and actions according to logic or given rules
- PC11.** Apply statistical techniques for data presentation
- PC12.** Use a range of textual and graphical methods to inform others
- PC13.** Communicate scientific information appropriately for the audience or recipient
- PC14.** Prepare and give formal and informal presentations
- PC15.** Seek, accept and, as appropriate, build on feedback and constructive criticism

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of communication
- KU2.** Principles for human interactivity



## Qualification Pack

- KU3.** The impact of one's own work on others, especially where related to diversity and equality
- KU4.** The specialist terminology associated with the work role and sector
- KU5.** The intention and purpose of statistical methods for data presentation
- KU6.** The limitations in reporting results
- KU7.** The uses of ICT, management information systems, and databases in chemical environments

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of EHS protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ EHS Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Communication and interpersonal skills</i>                                                                                                                                                                                                                                                                                | <b>30</b>    | <b>70</b>       | -             | -          |
| <b>PC1.</b> Establish and maintain interpersonal relations                                                                                                                                                                                                                                                                   | -            | -               | -             | -          |
| <b>PC2.</b> Work and interact with others including within teams                                                                                                                                                                                                                                                             | -            | -               | -             | -          |
| <b>PC3.</b> Provide technical support to chemists or other specialists                                                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC4.</b> Communicate using a full range of techniques for speaking, writing, body language, and active listening, for formal and informal purposes                                                                                                                                                                        | -            | -               | -             | -          |
| <b>PC5.</b> Use specialist terminology including where these are in another language                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC6.</b> Obtain information from all relevant sources, citing sources as required                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC7.</b> <ul style="list-style-type: none"> <li>• Read and apply the content of technical documents relating to               <ul style="list-style-type: none"> <li>• ◦ Analyses</li> <li>• ◦ Formulations</li> <li>• ◦ Procedural instructions</li> <li>• ◦ Specifications</li> <li>• ◦ Diagrams</li> </ul> </li> </ul> | -            | -               | -             | -          |
| <b>PC8.</b> Listen actively, asking questions appropriately for full understanding                                                                                                                                                                                                                                           | -            | -               | -             | -          |
| <b>PC9.</b> Use laboratory information and laboratory management systems, both digital and paper based                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC10.</b> Order information and actions according to logic or given rules                                                                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC11.</b> Apply statistical techniques for data presentation                                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <b>PC12.</b> Use a range of textual and graphical methods to inform others                                                                                                                                                                                                                                                   | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Communicate scientific information appropriately for the audience or recipient | -            | -               | -             | -          |
| <b>PC14.</b> Prepare and give formal and informal presentations                             | -            | -               | -             | -          |
| <b>PC15.</b> Seek, accept and, as appropriate, build on feedback and constructive criticism | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                            | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0355                                                               |
| <b>NOS Name</b>            | Communication and interpersonal skills                                  |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 1.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |

## Qualification Pack

### LFS/N0356: Techniques, procedures, and methods

#### Description

This NOS is about Techniques, procedures, and methods used by Chemical laboratory Technology

#### Scope

The scope covers the following :

- The scope of this qualification is about Techniques, procedures, and methods utilized by Chemical Laboratory Technology

#### Elements and Performance Criteria

##### *Techniques, procedures, and methods*

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare for laboratory tasks using the appropriate scientific techniques, procedures, and methods
- PC2.** Use specified instrumentation and laboratory equipment, including calibration as required
- PC3.** Evaluate the quality of materials or products to be used
- PC4.** Design or fabricate experimental apparatus to develop new products or processes
- PC5.** Perform laboratory tasks using specified methodologies, including standard operating procedures
- PC6.** Perform specific sampling duties including preparation and processing of samples as well as separation processes for mixtures of liquids and solids
- PC7.**
  - Perform cleaning and concentration processes such as
    - Distilling
    - Extracting
    - Evaporating
    - Chromatography
    - Potentiometry
    - Conductometry
- PC8.** Use titrimetric, volumetric, and gravimetric methods
- PC9.**
  - Use instrumental and electroanalytical methods such as
    - Photometry
    - Chromatography
    - Potentiometry
    - Conductometry
    - Electrophoresis
- PC10.**
  - Set up and conduct experiments, extractions, tests, and analyses, using techniques such as
    - Chromatography
    - Spectroscopy
    - Physical or chemical separation techniques
    - Microscopy
    - Electrophoresis
- PC11.** Determine the structures of organic and inorganic compounds

## Qualification Pack

- PC12.** Use synthesis techniques for organic, inorganic and polymer synthesis
- PC13.** Prepare chemical solutions for products or processes, following standardized formulae, or create experimental formulae
- PC14.** Take account of the need for validation of analytical procedures, methods, and instrumentation, including the use of suitable sampling methods.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The foundations of inorganic chemistry relative to structure and bonding
- KU2.** The chemistry of important elements and compounds
- KU3.** The principles and practical techniques of organic chemistry
- KU4.** Reaction mechanisms and functional group conversions
- KU5.** Concepts and practical techniques in physical chemistry, including thermodynamics, reaction kinetics, conductivity, electrochemical cells, and electrolysis
- KU6.** Principles of laboratory techniques and scientific experimentation
- KU7.** Principles of project management, and how these apply to laboratory work
- KU8.** The requirements for the development and validation of analytical methods and instrumentation, including understanding suitable sampling methods
- KU9.** Trends in support for experiments, including the use of kits

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of EHS protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ EHS Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Techniques, procedures, and methods</i>                                                                                                                                                                                                                                                                                  | 30           | 70              | -             | -          |
| <b>PC1.</b> Prepare for laboratory tasks using the appropriate scientific techniques, procedures, and methods                                                                                                                                                                                                               | -            | -               | -             | -          |
| <b>PC2.</b> Use specified instrumentation and laboratory equipment, including calibration as required                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> Evaluate the quality of materials or products to be used                                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <b>PC4.</b> Design or fabricate experimental apparatus to develop new products or processes                                                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC5.</b> Perform laboratory tasks using specified methodologies, including standard operating procedures                                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC6.</b> Perform specific sampling duties including preparation and processing of samples as well as separation processes for mixtures of liquids and solids                                                                                                                                                             | -            | -               | -             | -          |
| <b>PC7.</b> <ul style="list-style-type: none"> <li>• Perform cleaning and concentration processes such as               <ul style="list-style-type: none"> <li>• Distilling</li> <li>• Extracting</li> <li>• Evaporating</li> <li>• Chromatography</li> <li>• Potentiometry</li> <li>• Conductometry</li> </ul> </li> </ul> | -            | -               | -             | -          |
| <b>PC8.</b> Use titrimetric, volumetric, and gravimetric methods                                                                                                                                                                                                                                                            | -            | -               | -             | -          |
| <b>PC9.</b> <ul style="list-style-type: none"> <li>• Use instrumental and electroanalytical methods such as               <ul style="list-style-type: none"> <li>• Photometry</li> <li>• Chromatography</li> <li>• Potentiometry</li> <li>• Conductometry</li> <li>• Electrophoresis</li> </ul> </li> </ul>                 | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b><br>• Set up and conduct experiments, extractions, tests, and analyses, using techniques such as<br>• ◦ Chromatography<br>• ◦ Spectroscopy<br>• ◦ Physical or chemical separation techniques<br>• ◦ Microscopy<br>• ◦ Electrophoresis | -            | -               | -             | -          |
| <b>PC11.</b> Determine the structures of organic and inorganic compounds                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC12.</b> Use synthesis techniques for organic, inorganic and polymer synthesis                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC13.</b> Prepare chemical solutions for products or processes, following standardized formulae, or create experimental formulae                                                                                                               | -            | -               | -             | -          |
| <b>PC14.</b> Take account of the need for validation of analytical procedures, methods, and instrumentation, including the use of suitable sampling methods.                                                                                      | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                  | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0356                                                               |
| <b>NOS Name</b>            | Techniques, procedures, and methods                                     |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 7.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |



## Qualification Pack

### LFS/N0357: Data processing and record keeping

#### Description

This NOS is about Data processing and record keeping used by Chemical laboratory Technology

#### Scope

The scope covers the following :

- The scope of this qualification is about Data processing and record keeping utilized by Chemical Laboratory Technology

#### Elements and Performance Criteria

##### *Data processing and record keeping*

To be competent, the user/individual on the job must be able to:

- PC1.** Log and document laboratory work, including by using given house style, IT and statistical methods
- PC2.** Process and collate digital information from automated digital machines
- PC3.** Produce reliable, accurate data
- PC4.** Present the results of laboratory work and problem solving clearly and concisely in written and oral form
- PC5.** Write technical reports, using graphs and charts as appropriate
- PC6.** Check own work for codification, categorization, calculations, tabulations, and completeness
- PC7.** Acknowledge errors, inaccuracies, and shortcomings promptly
- PC8.** Arrange for information or data to be verified or audited
- PC9.** Archive documentation

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The rules relating to record keeping, traceability and confidentiality
- KU2.** Procedures for maintaining the security of records, in all forms used
- KU3.** The capabilities of software for recording and displaying data
- KU4.** Processes for ensuring the accuracy of information
- KU5.** The implications of error and inaccuracy
- KU6.** The required methods for referencing and citations

#### Generic Skills (GS)

User/individual on the job needs to know how to:



## Qualification Pack

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Data processing and record keeping</i>                                                                             | <b>30</b>    | <b>70</b>       | -             | -          |
| <b>PC1.</b> Log and document laboratory work, including by using given house style, IT and statistical methods        | -            | -               | -             | -          |
| <b>PC2.</b> Process and collate digital information from automated digital machines                                   | -            | -               | -             | -          |
| <b>PC3.</b> Produce reliable, accurate data                                                                           | -            | -               | -             | -          |
| <b>PC4.</b> Present the results of laboratory work and problem solving clearly and concisely in written and oral form | -            | -               | -             | -          |
| <b>PC5.</b> Write technical reports, using graphs and charts as appropriate                                           | -            | -               | -             | -          |
| <b>PC6.</b> Check own work for codification, categorization, calculations, tabulations, and completeness              | -            | -               | -             | -          |
| <b>PC7.</b> Acknowledge errors, inaccuracies, and shortcomings promptly                                               | -            | -               | -             | -          |
| <b>PC8.</b> Arrange for information or data to be verified or audited                                                 | -            | -               | -             | -          |
| <b>PC9.</b> Archive documentation                                                                                     | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                      | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0357                                                               |
| <b>NOS Name</b>            | Data processing and record keeping                                      |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 2.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |



## Qualification Pack

### LFS/N0358: Analysis, interpretation, and evaluation

#### Description

This NOS is about Analysis, interpretation, and evaluation used by Chemical laboratory Technology

#### Scope

The scope covers the following :

- The scope of this qualification is about Analysis, interpretation, and evaluation utilized by Chemical Laboratory Technology

#### Elements and Performance Criteria

##### *Data processing and record keeping*

To be competent, the user/individual on the job must be able to:

- PC1.** Maintain efficient kinaesthetic and fine motor skills
- PC2.** Apply personal techniques for sustained attentiveness and focus
- PC3.** Follow procedures to meet the workplace's quality standards
- PC4.** Analyse, interpret, and evaluate data and identify results requiring further investigation
- PC5.** Evaluate information to determine compliance with standards
- PC6.** Work autonomously within the parameters of the work role
- PC7.** Identify the meaning of outputs from the analytical techniques used and assess their importance
- PC8.** Use correct computational, statistical, and mathematical methods or formulae to solve problems
- PC9.** Identify by analysis the underlying principles, reasons, or facts determining results

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of quality management
- KU2.** Applications of quality management to production processes
- KU3.** Mathematical and statistical techniques used in analysis of scientific data
- KU4.** The nature, probabilities, sources, and types of errors
- KU5.** Principles and methods for quality control
- KU6.** Principles and applications for continuous improvement
- KU7.** The physiological implications of the work role

#### Generic Skills (GS)

User/individual on the job needs to know how to:





## Qualification Pack

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Data processing and record keeping</i>                                                                   | 30           | 70              | -             | -          |
| <b>PC1.</b> Maintain efficient kinaesthetic and fine motor skills                                           | -            | -               | -             | -          |
| <b>PC2.</b> Apply personal techniques for sustained attentiveness and focus                                 | -            | -               | -             | -          |
| <b>PC3.</b> Follow procedures to meet the workplace's quality standards                                     | -            | -               | -             | -          |
| <b>PC4.</b> Analyse, interpret, and evaluate data and identify results requiring further investigation      | -            | -               | -             | -          |
| <b>PC5.</b> Evaluate information to determine compliance with standards                                     | -            | -               | -             | -          |
| <b>PC6.</b> Work autonomously within the parameters of the work role                                        | -            | -               | -             | -          |
| <b>PC7.</b> Identify the meaning of outputs from the analytical techniques used and assess their importance | -            | -               | -             | -          |
| <b>PC8.</b> Use correct computational, statistical, and mathematical methods or formulae to solve problems  | -            | -               | -             | -          |
| <b>PC9.</b> Identify by analysis the underlying principles, reasons, or facts determining results           | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                            | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0358                                                               |
| <b>NOS Name</b>            | Analysis, interpretation, and evaluation                                |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 4.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |



## Qualification Pack

### LFS/N0359: Problem solving through the application of scientific methods

#### Description

This NOS is about Problem solving through the application of scientific methods used by Chemical laboratory Technology

#### Scope

The scope covers the following :

- The scope of this qualification is about Problem solving through the application of scientific methods utilized by Chemical Laboratory Technology

#### Elements and Performance Criteria

##### *Problem solving through the application of scientific methods*

To be competent, the user/individual on the job must be able to:

- PC1.** Recognize when there is either a problem or the likelihood of a problem
- PC2.** Identify or detect patterns within larger and distracting material
- PC3.** Apply suitable scientific methods to identify causes and achieve solutions
- PC4.**
  - Use logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems by, for example:
  - Applying general rules to specific issues to produce reasonable conclusions
  - Combining pieces of information to form reasonable conclusions or rules
- PC5.** Use creative thinking and problem solving to challenge assumptions, innovate, make new proposals, and build on existing ideas
- PC6.** Seek advice from senior colleagues as appropriate
- PC7.** Make recommendations for improved workflows or scientific solutions
- PC8.** Support new investigations and follow-up experiments for routine and non-routine analytical tasks
- PC9.** Take responsibility for personal development, demonstrating commitment to learning and self-improvement

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles and applications of scientific rules and methods to solve problems
- KU2.** Principles for critical thinking and complex problem solving
- KU3.** The scope and limits of their role and one's understanding and expertise in relation to problem-solving

#### Generic Skills (GS)



## Qualification Pack

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Problem solving through the application of scientific methods</i>                                                                                                                                                                                                                                                                                                                        | <b>30</b>    | <b>70</b>       | -             | -          |
| <b>PC1.</b> Recognize when there is either a problem or the likelihood of a problem                                                                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC2.</b> Identify or detect patterns within larger and distracting material                                                                                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <b>PC3.</b> Apply suitable scientific methods to identify causes and achieve solutions                                                                                                                                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC4.</b> <ul style="list-style-type: none"> <li>• Use logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems by, for example:</li> <li>• Applying general rules to specific issues to produce reasonable conclusions</li> <li>• Combining pieces of information to form reasonable conclusions or rules</li> </ul> | -            | -               | -             | -          |
| <b>PC5.</b> Use creative thinking and problem solving to challenge assumptions, innovate, make new proposals, and build on existing ideas                                                                                                                                                                                                                                                   | -            | -               | -             | -          |
| <b>PC6.</b> Seek advice from senior colleagues as appropriate                                                                                                                                                                                                                                                                                                                               | -            | -               | -             | -          |
| <b>PC7.</b> Make recommendations for improved workflows or scientific solutions                                                                                                                                                                                                                                                                                                             | -            | -               | -             | -          |
| <b>PC8.</b> Support new investigations and follow-up experiments for routine and non-routine analytical tasks                                                                                                                                                                                                                                                                               | -            | -               | -             | -          |
| <b>PC9.</b> Take responsibility for personal development, demonstrating commitment to learning and self-improvement                                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0359                                                               |
| <b>NOS Name</b>            | Problem solving through the application of scientific methods           |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 1.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |



## Qualification Pack

### LFS/N0360: Trends in applied Chemistry

#### Description

This NOS is about Trends in applied Chemistry used by Chemical laboratory Technology

#### Scope

The scope covers the following :

- The scope of this qualification is about Trends in applied Chemistry utilized by Chemical Laboratory Technology

#### Elements and Performance Criteria

##### *Trends in applied Chemistry*

To be competent, the user/individual on the job must be able to:

- PC1.** Install, commission and test automated laboratory systems
- PC2.** Install and configure programs
- PC3.** Develop simple programs
- PC4.** Switch on, switch off and operate automated laboratory systems
- PC5.** Optimize and implement adjustments and changes to automated laboratory systems
- PC6.** Maintain and service automated laboratory systems
- PC7.** Systematically search, localize, and eliminate faults, defects, and malfunctions on automated laboratory systems
- PC8.** Respond appropriately to change and change management processes

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The interdisciplinary nature of science
- KU2.** The role of applied chemistry in scientific developments
- KU3.** The growing impact of digitization
- KU4.** The growing importance of sustainability
- KU5.** New ethical concerns derived from new possibilities

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages





### Qualification Pack

- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and regulatory guidelines in the language prescribed by the company's SOP
- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules and regulatory guideline to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ Incharge or any other concerned authority clearly for escalating any emergency situation or hazard
- GS9.** Identify and resolve the issues and challenges

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Trends in applied Chemistry</i>                                                                                           | <b>30</b>    | <b>70</b>       | -             | -          |
| <b>PC1.</b> Install, commission and test automated laboratory systems                                                        | -            | -               | -             | -          |
| <b>PC2.</b> Install and configure programs                                                                                   | -            | -               | -             | -          |
| <b>PC3.</b> Develop simple programs                                                                                          | -            | -               | -             | -          |
| <b>PC4.</b> Switch on, switch off and operate automated laboratory systems                                                   | -            | -               | -             | -          |
| <b>PC5.</b> Optimize and implement adjustments and changes to automated laboratory systems                                   | -            | -               | -             | -          |
| <b>PC6.</b> Maintain and service automated laboratory systems                                                                | -            | -               | -             | -          |
| <b>PC7.</b> Systematically search, localize, and eliminate faults, defects, and malfunctions on automated laboratory systems | -            | -               | -             | -          |
| <b>PC8.</b> Respond appropriately to change and change management processes                                                  | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                             | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | LFS/N0360                                                               |
| <b>NOS Name</b>            | Trends in applied Chemistry                                             |
| <b>Sector</b>              | Life Sciences                                                           |
| <b>Sub-Sector</b>          | Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology |
| <b>Occupation</b>          | Quality                                                                 |
| <b>NSQF Level</b>          | 4.5                                                                     |
| <b>Credits</b>             | 1.0                                                                     |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                              |
| <b>Next Review Date</b>    | 08/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                              |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Assessment is governed by the WorldSkills Assessment Strategy. The Strategy establishes the principles and techniques to which WorldSkills assessment and marking must conform.
2. Expert assessment practice lies at the heart of the WorldSkills Competition. For this reason, it is the subject of continuing professional development and scrutiny. The growth of expertise in assessment will inform the future use and direction of the main assessment instruments used by the WorldSkills Competition: the Marking Scheme, Test Project, and Competition Information
3. System (CIS).
4. Assessment at the WorldSkills Competition falls into two broad types: Measurement and Judgement. For both types of assessment, the use of explicit benchmarks against which to assess each Aspect is essential to guarantee quality.
5. The Marking Scheme must follow the weightings within the Standards. The Test Project is the assessment vehicle for the skill competition, and therefore also follows the Standards. The CIS enables the timely and accurate recording of marks; its capacity for scrutiny, support, and feedback is continuously



## Qualification Pack

expanding.

6. The Marking Scheme, in outline, will lead the process of Test Project design. After this, the Marking Scheme and Test Project will be designed, developed, and verified through an iterative process, to ensure that both together optimize their relationship with the Standards and the Assessment

7. Strategy. They will be agreed by the Experts and submitted to WSI for approval together, to demonstrate their quality and conformity with the Standards.

8. Prior to submission for approval to WSI, the Marking Scheme and Test Project will liaise with the WSI Skill Advisors for quality assurance and to benefit from the capabilities of the CIS.

### Minimum Aggregate Passing % at QP Level : 70

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Minimum Passing % at NOS Level: 70

**(Please note:** A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

## Assessment Weightage

### Compulsory NOS

| National Occupational Standards                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|-------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| LFS/N0354.Work organization and management                              | 30           | 70              | -             | -          | 100         | 10        |
| LFS/N0355.Communication and interpersonal skills                        | 30           | 70              | -             | -          | 100         | 10        |
| LFS/N0356.Techniques, procedures, and methods                           | 30           | 70              | -             | -          | 100         | 35        |
| LFS/N0357.Data processing and record keeping                            | 30           | 70              | -             | -          | 100         | 10        |
| LFS/N0358.Analysis, interpretation, and evaluation                      | 30           | 70              | -             | -          | 100         | 15        |
| LFS/N0359.Problem solving through the application of scientific methods | 30           | 70              | -             | -          | 100         | 10        |



### Qualification Pack

| National Occupational Standards       | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|---------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| LFS/N0360.Trends in applied Chemistry | 30           | 70              | -             | -          | 100         | 10         |
| <b>Total</b>                          | <b>210</b>   | <b>490</b>      | <b>-</b>      | <b>-</b>   | <b>700</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |
| <b>IT</b>   | Information technology                          |
| <b>NOS</b>  | National Occupational Standards                 |
| <b>HPLC</b> | High Performance Liquid Chromatography          |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |